

IAN SEQUEIRA

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EDUCATION

University of California, Santa Barbara

Santa Barbara, CA

B.S. Statistics & Data Science, B.A. Economics

Sept 2023 – June 2027

- GPA: 3.82 / 4.0 | Dean's Honors (6 quarters)
- Coursework: Machine Learning, Design of Experiments, Stochastic Processes, Regression Analysis, Econometrics, Probability & Statistics, Proof Writing

RESEARCH EXPERIENCE

UCSB Department of Statistics & Applied Probability

Santa Barbara, CA

Undergraduate Researcher – Prof. Annie Qu & Prof. Xiaowu Dai

Apr 2026 – Present

- Developing the **Compositional Coordination Confidence** framework, which provides anytime valid confidence guarantees for multi-agent LLM pipelines via typed e-processes that compose across DAG edges
- Proving composition theorems; Monte Carlo simulations validated the framework across sequential, fan-in, and fan-out agent topologies

UC Irvine Summer Institute for Biostatistics

Irvine, CA

Research Fellow – Prof. Dan Gillen & Dr. Josh Grill

June 2025 – Aug 2025

- Constructed GLM, ridge, and CART classifiers in R identifying Alzheimer's biomarkers (pTau217, PACC, ADL) across 1,150+ A4 trial participants (0.835 AUC)
- Developed sample size reduction framework achieving 80% power with 188 vs. 500+ participants
- Co-author on biomarker-guided trial design study; research presented at *CTAD 2025*, San Diego; manuscript in preparation

WORK EXPERIENCE

Alida Inc.

Remote

Student Intern Data Engineer

Sept 2024 – Present

- Built multi-mode AI data exploration platform over 47 healthcare datasets using multi-agent routing, graph analysis, and automated SQL generation in Databricks
- Extended platform with spec-driven ML prediction pipeline featuring typed contracts, dual-layer verification, and fault-tolerant LLM orchestration with AutoML training

PROJECTS

Healthcare Financial Toxicity Prediction

Mar 2026

R, XGBoost, BART, SHAP, fairmodels

- PSTAT 231 (graduate Statistical ML) final project: predicted catastrophic out-of-pocket burden on 20,962 MEPS respondents; benchmarked 6 classifiers under 10-fold CV (XGBoost ROC-AUC 0.874)
- Audited subgroup fairness with DALEX and produced SHAP-based odds ratio interpretations

Research Advisor Finder

Nov 2025 – Feb 2026

FastAPI, PostgreSQL, pgvector, Next.js, OpenAI, Claude API

- Built full stack faculty discovery platform matching student research interests to 15k+ professors across 12 STEM fields via hybrid semantic search and CV-driven interest extraction

LEADERSHIP AND ACTIVITIES

Project Executive, ACM Industry, UCSB

Sept 2025 – Present

- Leading cross-functional student teams delivering applied data science and software solutions for industry partner Turing

Finance Committee, CampMed, UCSB

Sept 2025 – Present

- Coordinating fundraising and event logistics for health equity outreach to underrepresented high school students

TECHNICAL SKILLS

Programming: R, Python, SQL, C++, JavaScript/TypeScript, HTML & CSS, LaTeX

Frameworks & Tools: tidymodels, xgboost, rstan, scikit-learn, PyTorch, SHAP, Databricks, LangGraph

Languages: English, Spanish | **Memberships:** ASA, IMS, UCSB Innovators Circle